

# CHEMICAL REACTIONS AND EQUATIONS

## CHEMICAL REACTIONS

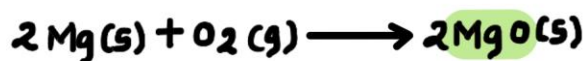
- Process in which new substance/products are formed

e.g., Hydrogen + oxygen  $\longrightarrow$  Water (New)

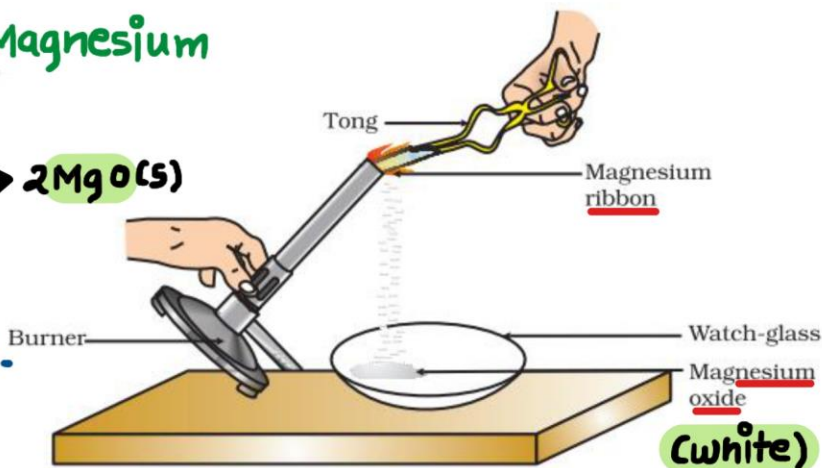
Reactants

Product

e.g., Burning of Magnesium in air



- NOTE:** Mg-ribbon cleaned with sand-paper before burning



- In a chemical reaction:

- gas can be evolved
- colour change
- State change; (s)  $\rightarrow$  (l)
- Temperature change
- Precipitate (insoluble) can be formed  
 $\hookrightarrow \text{BaSO}_4$  (white ppt.)

e.g., Burning of paper ; Fermentation of grape ; Souring of milk ; Rusting of Iron ; Digestion of food ; Respiration ; Baking cake ; chemical battery



## CHEMICAL EQUATIONS

- Simple representation of a reaction with symbols & formulas



## BALANCING A CHEMICAL EQUATIONS

- No. of atoms of each element should remain same before & after a reaction due to law of conservation of mass





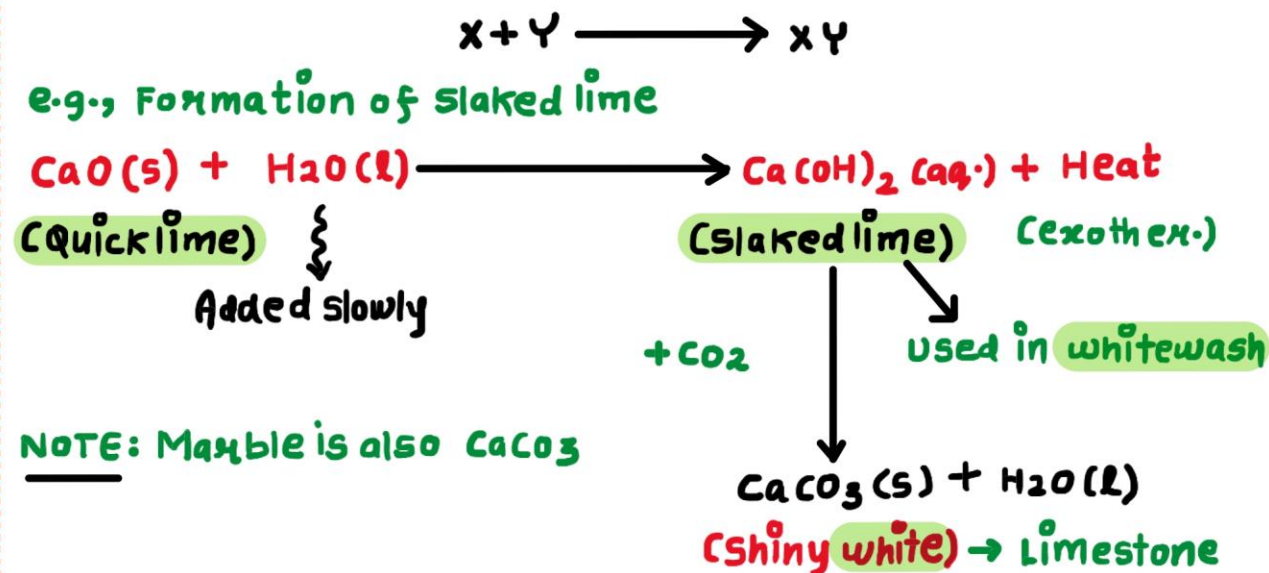
|                                                                             |                                                                                    |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Write an <b>unbalanced equation</b>                                         | $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$    |
| List the <b>number of atoms of different elements</b>                       | LHS/ RHS<br>Fe: 1/3<br>H: 2/2<br>O: 1/4                                            |
| Start balancing with the element having max. number of atoms (i.e., oxygen) | $\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$   |
| Balance other elements i.e., Fe and H in similar way                        | $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$ |
| Check whether balanced or not                                               | YES                                                                                |

| Unbalanced Equations                                                                   | Balanced Equations                                                                                                                                               |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Zn} + \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$                        | $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$                                                                                                 |
| $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$        | $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$                                                                               |
| $\text{BaCl}_2 + \text{Al}_2(\text{SO}_4)_3 \rightarrow \text{BaSO}_4 + \text{AlCl}_3$ | $3\text{BaCl}_2 + \text{Al}_2(\text{SO}_4)_3 \rightarrow 3\text{BaSO}_4 + 2\text{AlCl}_3$                                                                        |
| $\text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbO} + \text{NO}_2 + \text{O}_2$           | $\text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbO} + 2\text{NO}_2 + \text{O}_2$<br>$2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ |

## TYPES OF CHEMICAL REACTIONS

### COMBINATION REACTIONS

- 2 or more reactants combine to form a single product



| Other Examples                   | Chemical Equations                                                                          |
|----------------------------------|---------------------------------------------------------------------------------------------|
| Burning of Coal                  | $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$                      |
| Formation of Water               | $2\text{H}_2\text{(g)} + \text{O}_2 \rightarrow 2\text{H}_2\text{O(l)}$                     |
| Formation of Mg(OH) <sub>2</sub> | $\text{MgO(s)} + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2\text{(s)}$                  |
| Formation of CuO                 | $2\text{Cu(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{CuO(s)}$<br>(Shiny brown to Black) |

## DECOMPOSITION REACTION

- one reactant breaks into 2 or more products
- The decomposition can be **electrochemical** (use electricity); **photochemical** (use light) & **thermal** (use temp.)
- $\therefore$  endothermic

## THERMAL DECOMPOSITION

### 1 Heating Ferrous sulphate



Ferrioxide: Brown

Suffocating smell; acidic;  
pollutants; wafted gently

Taken in dry test tube  
(make green crystals)

### 2 Decomposition of $\text{CaCO}_3$



Limestone

Quicklime: cement making

### 3 Decomposition of Lead nitrate:

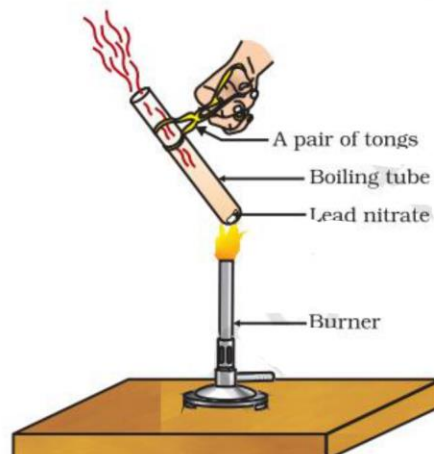
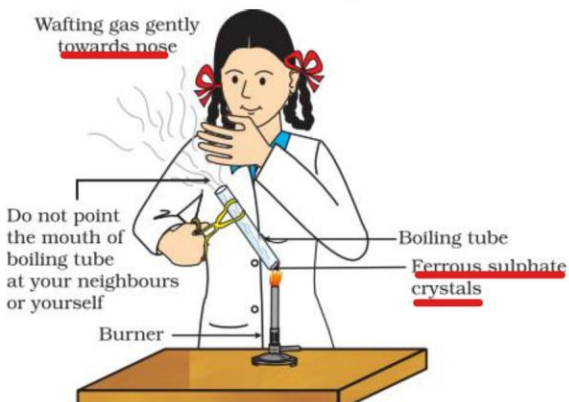


Pb-nitrate (white)

Yellow residue → Brown fumes  
acidic, irrit.

Heat with cracking sound

Wafting gas gently  
towards nose



## ELECTROLYSIS OF WATER



(not pure)

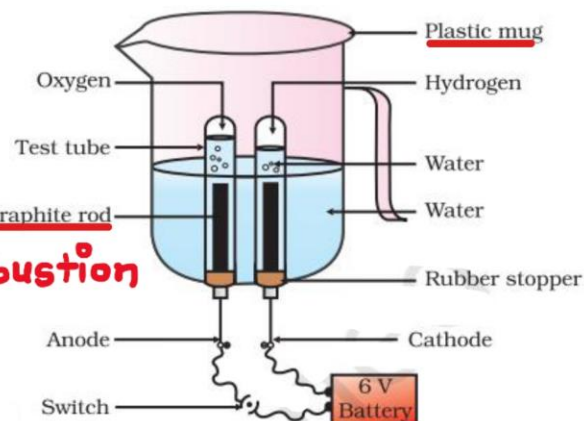
Burns with  
pop sound

Support combustion

• cation:  $\ominus$ ve; cathode:  $\ominus$ ve

• Anion:  $\ominus$ ve; anode:  $\oplus$ ve

$\nearrow \text{H}_2$   
 $\searrow \text{O}_2$



NOTE:  $\text{O}_2 : \text{H}_2 = 1 : 2$



## PHOTOLYSIS

•  $\text{AgCl}$  &  $\text{AgBr}$  are kept in black glass bottles & are used in white & black photography



white

grey

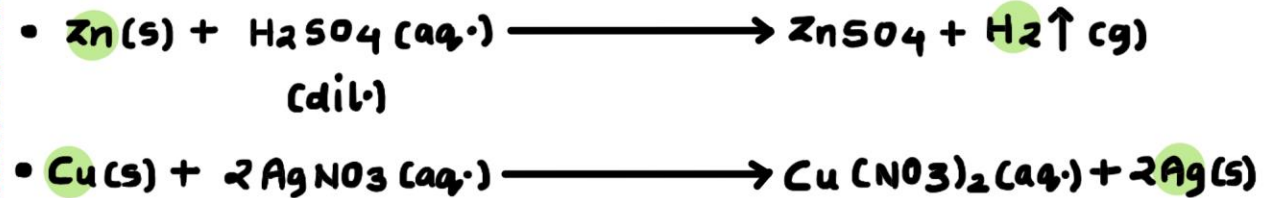
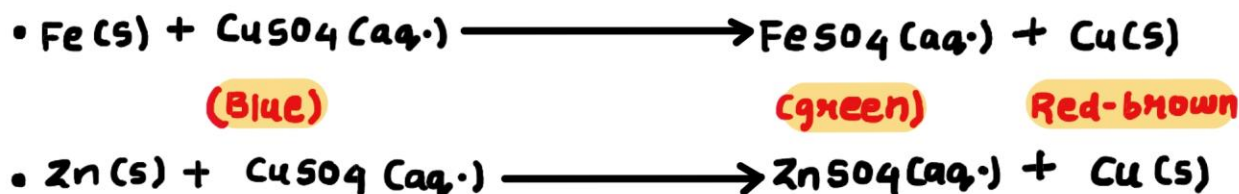
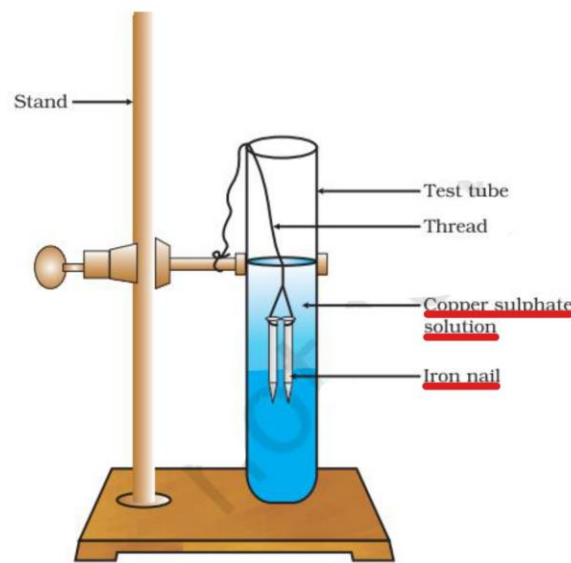


## DISPLACEMENT REACTION

• Here, more reactive element displaces less reactive element

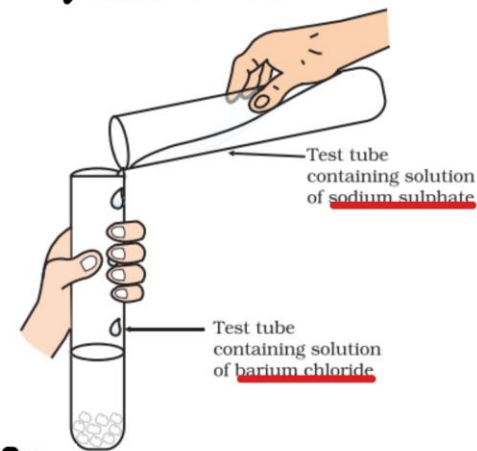
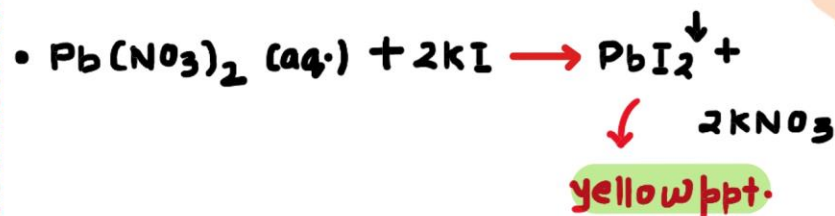
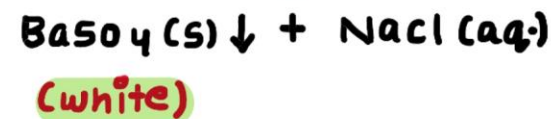
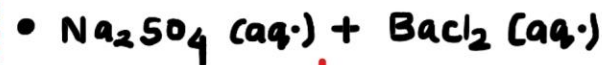


| More to Less Reactive | Trick          |
|-----------------------|----------------|
| K (Potassium)         | Kudi           |
| Na (Sodium)           | Naal           |
| Ca (Calcium)          | Car            |
| Mg (Magnesium)        | Maango         |
| Al (Aluminium)        | Alto           |
| Zn (Zinc)             | Zen            |
| Fe (Iron)             | Ferrari        |
| Pb (Lead)             | Pub            |
| H (Hydrogen)          | Ham            |
| Cu (Copper)           | Chale          |
| Hg (Mercury)          | Hug Kiya       |
| Ag (Silver)           | Silver Ring Di |
| Au (Gold)             | Gold Ring Di   |



## DOUBLE DISPLACEMENT REACTIONS

- Happens due to exchange of ions b/w reactants

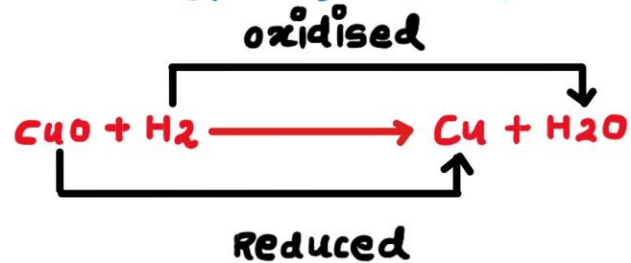


| Exothermic Reactions                                                                                | Endothermic Reactions                                                  |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Energy released ( <span style="color: red;">exit</span> )                                           | Energy needed ( <span style="color: red;">enter</span> )               |
| e.g., respiration; burning of natural gas; decomposition of vegetable compost; burning of Mg ribbon | All <span style="color: green;">thermal decomposition</span> reactions |



## REDOX REACTIONS

- **oxidation**: gain of oxygen; loss of hydrogen or electron
- **Reduction**: loss of oxygen; gain of hydrogen or electron



- **oxidising agent**: get reduced to oxidise other
- **Reducing agent**: get oxidised to reduce other



- e.g., oxidation of copper



## CORROSION

- Metal attacked moisture,  $\text{O}_2$ , gases, acids to make undesirable substance

e.g., Rusting of iron: **oxide (brown)**

Tarnishing of silver: **sulphate (black)**

Tarnishing of copper: **carbonate (green)**

Prevention: **Painting, oiling, galvanising (zinc coat)**

## RANCIDITY

- change in **Smell & odour** of fat/oil containing food due to oxidation
- **Prevention**: Refrigeration; adding antioxidants; **Nit. flushing**; vacuum tight packing

# ACID, BASES AND SALTS

## ACIDS AND BASES

| ACID                                                                                                                                                       | BASE                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Give $H^+$ (aq.) as only positive ion in aq. Solution (conduct electricity)                                                                                | Give $OH^-$ (aq.) as only negative ion in aq. Solution (conduct electricity)                                 |
| Sour in taste<br>Corrosive in nature                                                                                                                       | Bitter in taste<br>Soapy in touch                                                                            |
| Turn Blue litmus <b>Red</b>                                                                                                                                | Turn Red litmus <b>Blue</b>                                                                                  |
| <b>Weak</b> (partially dissociates in water): $CH_3COOH$ , citric acid, lactic acid<br><b>Strong</b> (completely dissociates): $HCl$ , $H_2SO_4$ , $HNO_3$ | <b>Weak</b> : $Cu(OH)_2$ , $Fe(OH)_2$ , $NH_4OH$ , $Al(OH)_3$<br><b>Strong</b> : $NaOH$ , $KOH$ , $Ca(OH)_2$ |

- **Colour indicators**: colour change in acidic, basic medium

| Indicator                              | Acid       | Base             |
|----------------------------------------|------------|------------------|
| Red Litmus (natural)                   | -          | <b>Blue</b>      |
| Blue Litmus                            | <b>Red</b> | -                |
| Phenolphthalein (colourless-synthetic) | Colourless | <b>Pink (BP)</b> |
| Methyl Orange (synthetic)              | <b>Red</b> | Yellow           |

- **Olfactory indicators**: changes odour in acidic & basic med.  
e.g., onion, vanilla essence, clove oil: smell **vanishes** in base & no change in acids

## CHEMICAL PROPERTIES

- 1 **Reaction with water**: acid gives  $H^+$  ions in  $H_2O$



$\therefore H^+$  ions can't exist alone  $\therefore$  make **hydronium ion**



- Base gives  $OH^-$  ion in  $H_2O$



- Base soluble in  $H_2O$  are called **Alkali** like  $NaOH$ ,  $KOH$ ,  $NH_4OH$
- $Zn(OH)_2$ ,  $Fe(OH)_2$  are not soluble  $\therefore$  not alkali
- $Ca(OH)_2$ ,  $Mg(OH)_2$  are partially soluble

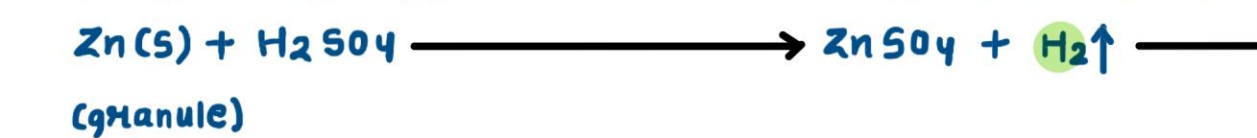
**NOTE**: Always add acids slowly to water  $\therefore$  exothermic

- 2 **Reaction of acid & Base i.e., neutralisation**

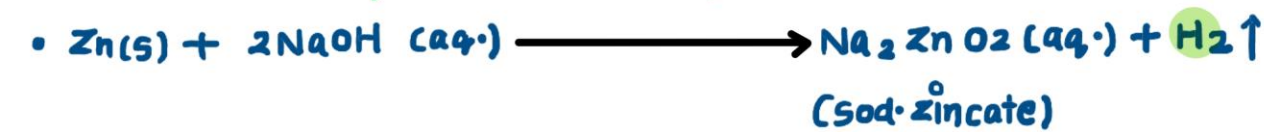




**3 Reaction with metals:** dil. acids react with metals to give  $\text{H}_2$  (g) & salt



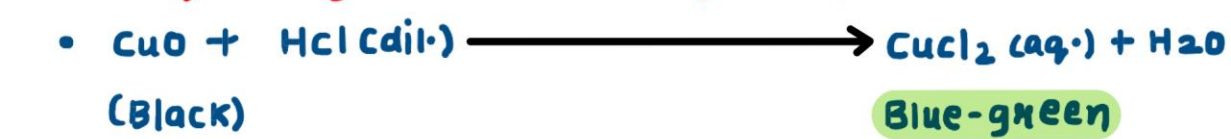
• **Base + metal give salt +  $\text{H}_2$  (g)**



**4 Reaction with metal oxides:**



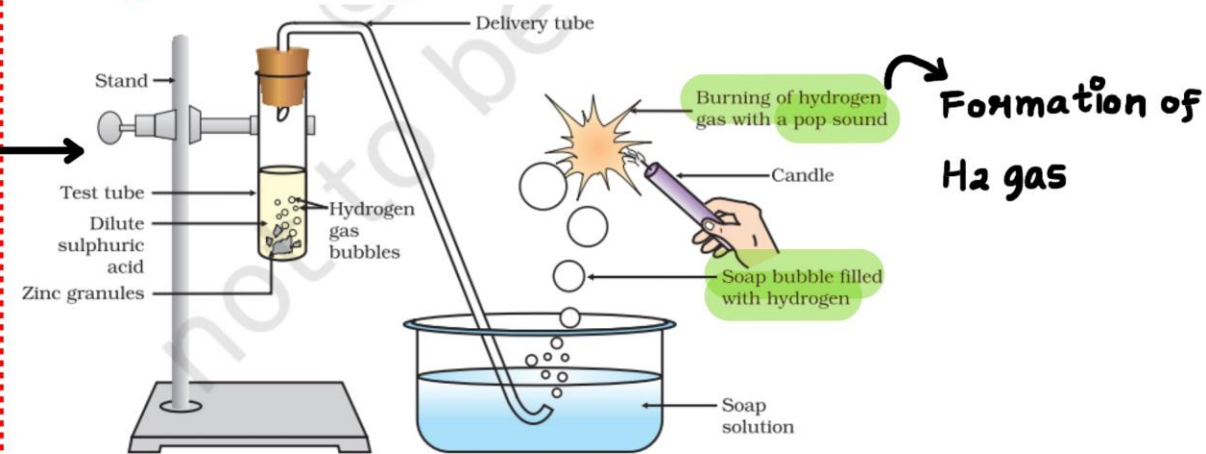
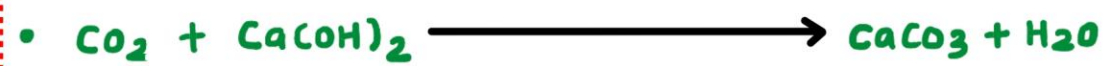
$\hookrightarrow \therefore$  generally basic i.e.,  $\text{ZnO}$ ,  $\text{MgO}$ ,  $\text{CaO}$ ,  $\text{FeO}$ ,  $\text{CuO}$



**5 Reaction with Non metal oxides**



$\hookrightarrow \therefore$  majorly acidic i.e.,  $\text{SO}_2$ ,  $\text{SO}_3$ ,  $\text{NO}_2$ ,  $\text{CO}_2$

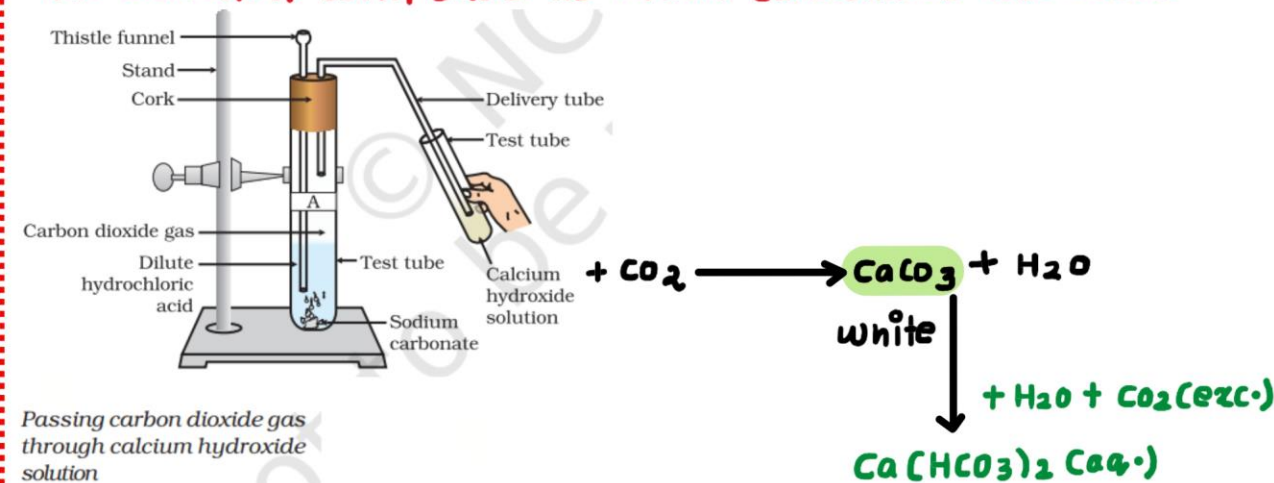


Reaction of zinc granules with dilute sulphuric acid and testing hydrogen gas by burning

**6 Reaction with Metal carbonates**



• **No reaction with base as metal carbonates are basic**

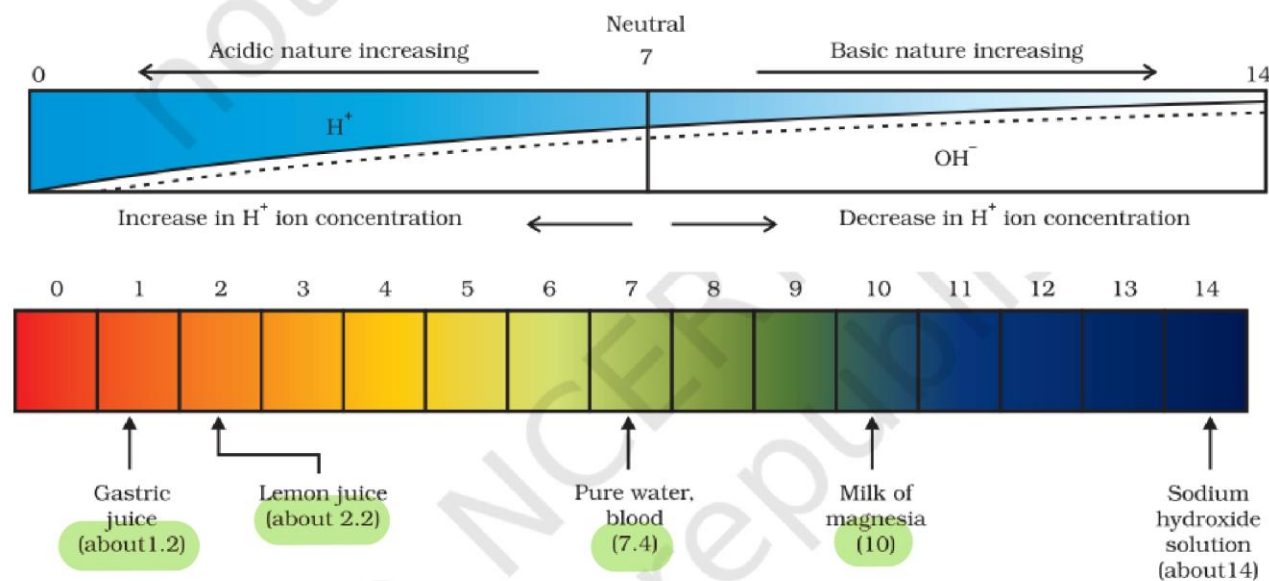


## 7 Reaction with Metal - hydrogen carbonate

- Metal hydrogen carbonate + acid  $\longrightarrow$  Salt +  $H_2O$  +  $CO_2 \uparrow$
- $NaHCO_3 (s) + HCl (aq.) \longrightarrow NaCl (aq.) + H_2O (l) + CO_2 (g)$
- Do not react with bases as metal-H-carbonates are basic

## STRENGTH OF ACIDS AND BASES

- pH is measure of  $H^+$  ions in a solution
- It ranges from 0  $\rightarrow$  14
- <7 for acids ; 7 is neutral ( $H_2O$ ) ; >7 for bases  
 $\swarrow H^+ \uparrow$   $\searrow H^+ \downarrow$
- Universal indicator: pH paper



## pH in DAILY LIFE

|                                                                                         |                                                                                                                    |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| pH of Human Body                                                                        | 7-7.8                                                                                                              |
| Acid rain (<5.6pH)                                                                      | Makes aquatic life difficult                                                                                       |
| HCl in stomach                                                                          | pH 1.2: its higher secretion can cause acidity which is neutralised by antacids like milk of magnesia- $Mg(OH)_2$  |
| Decay in tooth (calcium hydroxyapatite)                                                 | Bacteria make acids and make teeth pH less than 5.5; which is neutralised by base in toothpaste                    |
| Bee sting has formic acid which cause itching<br>Nettle leave sting: has methanoic acid | Treated by base like baking soda ( $NaHCO_3$ )<br>Treated by rubbing leaves of dock plant which grow nearby nettle |

## SOME NATURAL ACIDS

| Sources       | Acid          |
|---------------|---------------|
| Vinegar       | Acetic Acid   |
| Lemon/ Orange | Citric Acid   |
| Curd          | Lactic Acid   |
| Tomato        | Oxalic Acid   |
| Tamarind      | Tartaric Acid |

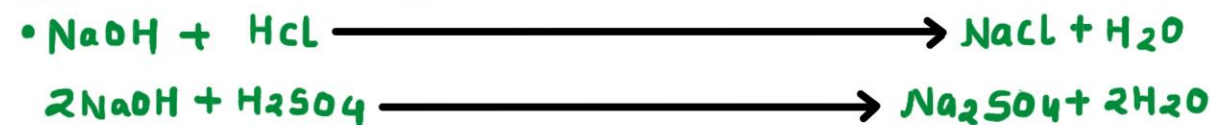


# SALTS

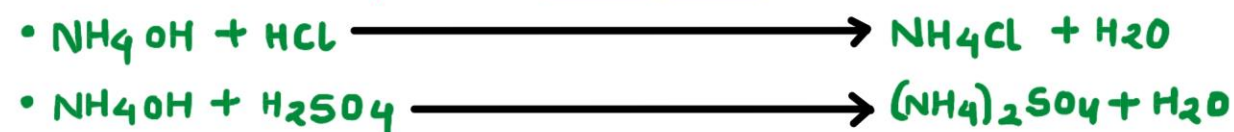
- Ionic compound produced by neutralisation of acid & base; crystalline solids & conducts electricity

## PH of salts:

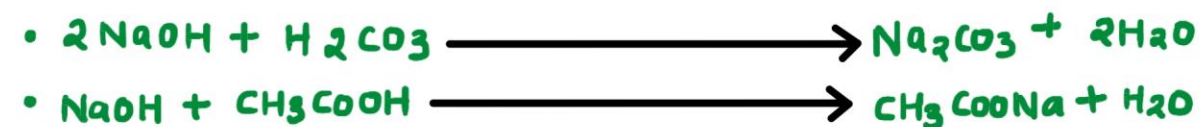
- 1 **Neutral salts (pH=7)**; salts of strong acid & str. base



- 2 **Acidic salts (pH<7)**; Strong acid + weak base



- 3 **Basic salts (pH>7)**; Strong base + weak acid



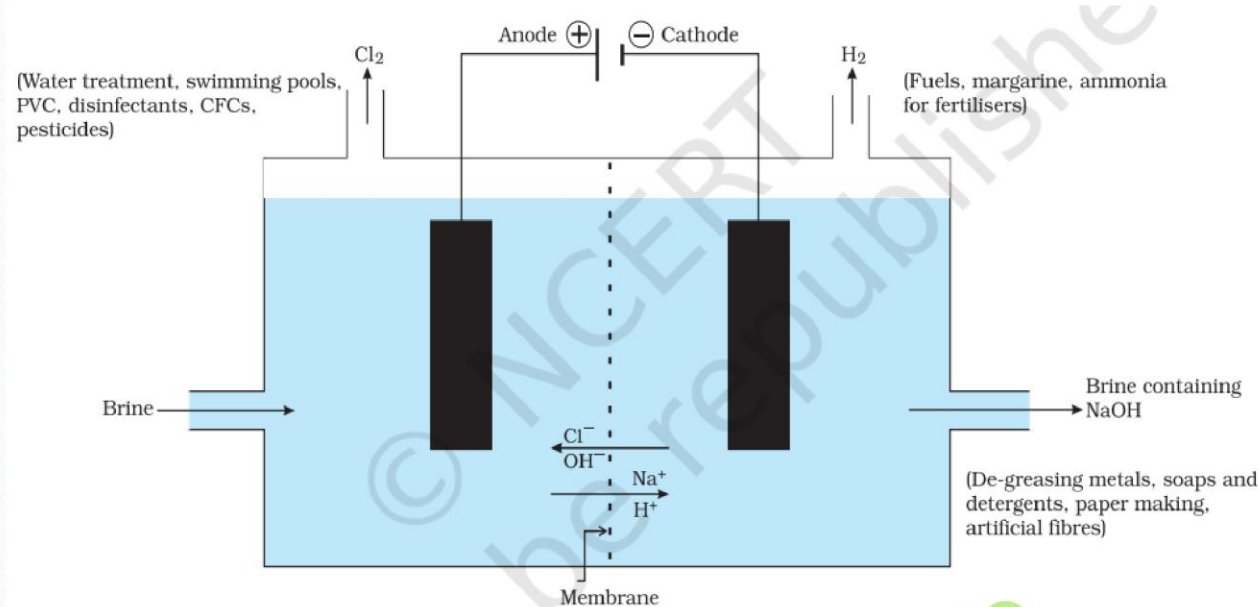
## CHEMICALS FROM COMMON SALT (NaCl)

- 1 **NaOH / caustic soda**

Preparation: By **chlor-alkali process**



↳ **Electrolysis of Brine**



**CATHODE: NaOH, H<sub>2</sub> ; ANODE (+ve): Cl<sub>2</sub>**

Soap, detergents,  
paper making

Fuel, NH<sub>3</sub>, fertiliser,  
Margarine

Disinfectant, CFCs,  
pesticides, PVCs

- 2 **Bleaching powder: CaOCl<sub>2</sub>**



(slaked lime) ↳ from last reaction

- Used for **bleaching** in textile, paper, laundry; as an ox. agent; sterilising H<sub>2</sub>O

- 3 **Baking soda: NaHCO<sub>3</sub>**



- It is a mild non-corrosive basic salt

- Preparation of Baking powder

- $\text{NaHCO}_3 + \text{H}^+$  from mild acid  $\longrightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Sod. salt of acid}$
- work as antacid ; help in baking cakes; used as fire exting.

4 Washing soda :  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

- $2\text{NaHCO}_3 \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$

- $\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \longrightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

- used in glass, soap & paper industry ; domestic cleanser ; remove permanent hardness of  $\text{H}_2\text{O}$

## WATER OF CRYSTALLISATION

- Fixed no. of  $\text{H}_2\text{O}$  molecules attached to one molecule of any salt

e.g.,  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  ;  $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$  ;  $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$  ;

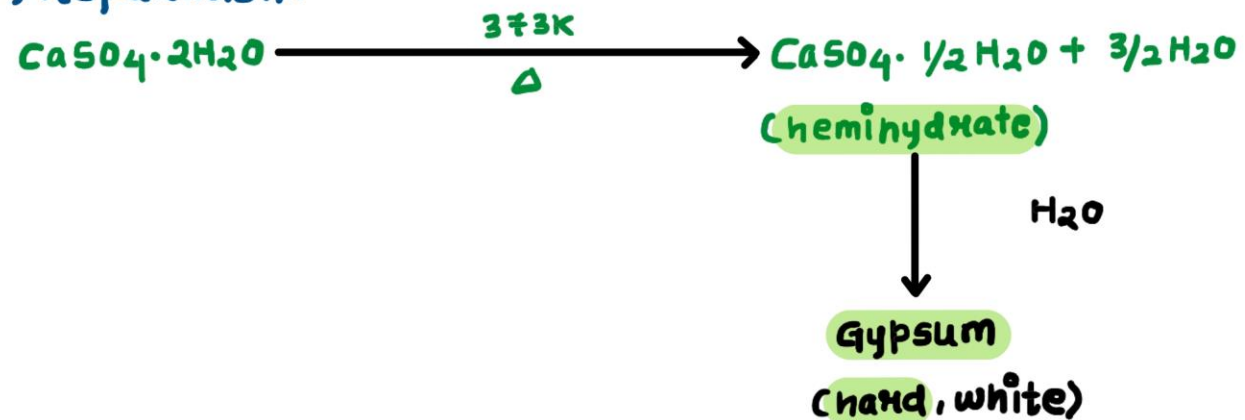
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  (gypsum)

- when crystals heated,  $\text{H}_2\text{O}$  gets removed

## PLASTER OF PARIS (POP)

- It is  $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$

Preparation:





# METALS AND NON-METALS

## PHYSICAL PROPERTIES

| Metals                                                                         | Non-Metals                                                                                                                                            |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>State:</b> generally solid at room temperature (except mercury and gallium) | Either solid (C, S) or gas (N <sub>2</sub> , Cl <sub>2</sub> , O <sub>2</sub> , F <sub>2</sub> ) at room temperature (except Bromine which is liquid) |
| <b>Lustrous/ shiny</b>                                                         | <b>Non-lustrous</b> (except graphite and iodine)                                                                                                      |
| <b>Generally hard</b> (except Li, Na, K which can be cut with knife)           | <b>Soft</b> (except diamond)                                                                                                                          |
| <b>Malleable</b> (can be drawn in sheets) and <b>ductile</b> (drawn in wires)  | Not malleable and ductile                                                                                                                             |
| <b>Good conductors</b> of electricity e.g., Au, Ag, Cu (Pb is poor conductor)  | Poor ability to conduct                                                                                                                               |
| <b>Conducts heat</b> e.g., Ag (mercury is poor conductor)                      | Poor ability to conduct                                                                                                                               |
| <b>High melting point</b> (except Caesium and Gallium which melts in palm)     | <b>Low melting point</b> (except some carbon alloys)                                                                                                  |
| <b>Sonorous</b> (produce sound on strike)                                      | <b>Non-sonorous</b>                                                                                                                                   |
| <b>Density:</b> generally high                                                 | <b>Low</b>                                                                                                                                            |

• Cu, Al: used to make cooking vessel (mpt ↑, conductor ↑)

## CHEMICAL PROPERTIES

|     |            |                |
|-----|------------|----------------|
| K   | Potassium  | Most reactive  |
| Na  | Sodium     |                |
| Ca  | Calcium    |                |
| Mg  | Magnesium  |                |
| Al  | Aluminium  |                |
| Zn  | Zinc       |                |
| Fe  | Iron       |                |
| Pb  | Lead       |                |
| [H] | [Hydrogen] |                |
| Cu  | Copper     |                |
| Hg  | Mercury    |                |
| Ag  | Silver     |                |
| Au  | Gold       | Least reactive |

Reactivity decreases

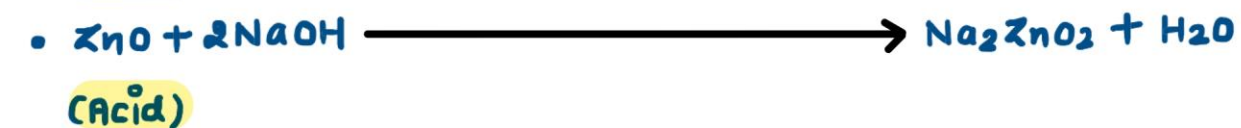
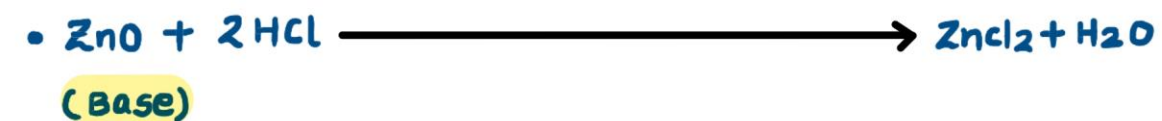
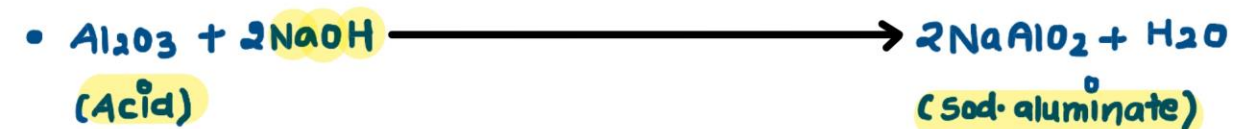
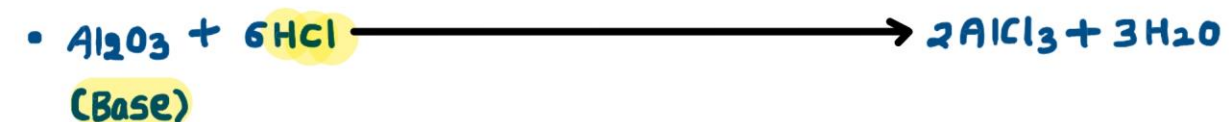
### 1 Reaction with air (O<sub>2</sub>)

- Metal + O<sub>2</sub> → Metal oxide (basic)
  - Na, K: react vigorously & catch fire if kept open  
∴ stored in kerosene oil
- $$4\text{Na} + \text{O}_2 \longrightarrow 2\text{Na}_2\text{O (s)}$$

- This  $\text{Na}_2\text{O}$  dissolve in  $\text{H}_2\text{O}$  to form alkali ( $\therefore$  alkali metals)



- $\text{Mg, Al, Zn, Pb}$ : React slowly & form a protective layer  
 $\downarrow$  form amphoteric oxides (behave like acid & base)  
 Burn with white dazzling flame



- $\text{Fe}$ : donot react directly but iron fillings can react

- $\text{Cu}$ : don't react at room temp. but react at  $\uparrow$  Temp. to form black coat of  $\text{CuO}$

- $\text{Ag, Au}$ : donot react even at Temp.  $\uparrow$

| Ion           | Flame Colour     |
|---------------|------------------|
| $\text{Li}^+$ | Crimson (LaaaLi) |
| $\text{Na}^+$ | Yellow (NaYaaaa) |
| $\text{K}^+$  | Lilac (KaLi)     |
| $\text{Ca}^+$ | Red (CaR)        |
| $\text{Cu}^+$ | Green            |

## 2 Reaction with acids:

Metals + dil. acids  $\longrightarrow$  Salt +  $\text{H}_2 \uparrow$  (popsound)



- $\text{H}_2$  is not liberated when metal reacts with  $\text{HNO}_3$  as it is a strong oxidising agent  $\therefore$   $\text{H}_2$  is converted to  $\text{H}_2\text{O}$  & it is itself reduced to  $\text{N}_2\text{O, NO, NO}_2$

Exception:  $\text{Mg}$  &  $\text{Mn}$  react with dil.  $\text{HNO}_3$  to give  $\text{H}_2$

- Reactivity & exothermic order:  $\text{Mg} > \text{Al} > \text{Zn} > \text{Fe}$

- $\text{Cu}$  &  $\text{Ag}$ : do not react with dil.  $\text{HCl}$

- Aqua Regia: Royal Solution: conc.  $\text{HCl}$  & conc.  $\text{HNO}_3$  in 3:1

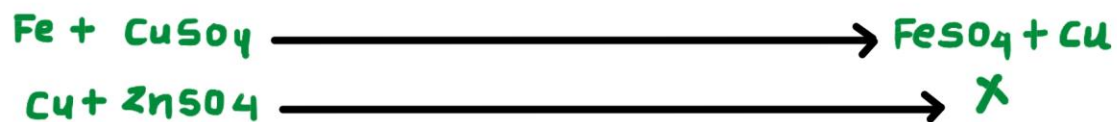


### 3 React with H<sub>2</sub>O

- Metal + H<sub>2</sub>O → Metal oxide + H<sub>2</sub>↑  
Metal oxide + H<sub>2</sub>O → Metal Hydroxide
- Na, K: react violently with cold H<sub>2</sub>O & catches fire  
 $\text{Na (s)} + 2\text{H}_2\text{O} \longrightarrow 2\text{NaOH (aq)} + \text{H}_2\uparrow$
- Ca: react less violently & start floating (H<sub>2</sub> bubble atta)  
 $\text{Ca (s)} + 2\text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 \text{ (aq)} + \text{H}_2$
- Mg: reacts with hot water & start to float
- Al, Zn, Fe: react with steam  
 $2\text{Al (s)} + 3\text{H}_2\text{O (g)} \longrightarrow \text{Al}_2\text{O}_3 \text{ (s)} + 3\text{H}_2 \text{ (g)}$   
 $3\text{Fe (s)} + 4\text{H}_2\text{O (g)} \longrightarrow \text{Fe}_3\text{O}_4 \text{ (s)} + 4\text{H}_2 \text{ (g)}$
- Pb, Cu, Ag, Au: do not react
- Note: K, Na, Ca, Mg can make hydroxides ∴ ↑ reactive

### 4 Reaction with other metals i.e., displacement

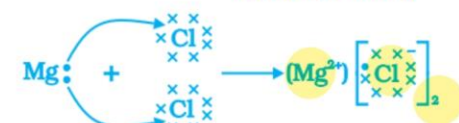
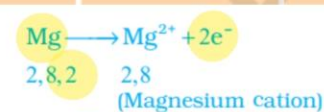
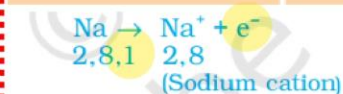
Here, ↑ reactive metal displaces the less reactive one



## IONIC/ ELECTROVALENT COMPOUNDS

- Metal + Non metal = Ionic compound

| Type of element | Element        | Atomic number | Number of electrons in shells |   |   |   |
|-----------------|----------------|---------------|-------------------------------|---|---|---|
|                 |                |               | K                             | L | M | N |
| Noble gases     | Helium (He)    | 2             | 2                             |   |   |   |
|                 | Neon (Ne)      | 10            | 2                             | 8 |   |   |
|                 | Argon (Ar)     | 18            | 2                             | 8 | 8 |   |
| Metals          | Sodium (Na)    | 11            | 2                             | 8 | 1 |   |
|                 | Magnesium (Mg) | 12            | 2                             | 8 | 2 |   |
|                 | Aluminium (Al) | 13            | 2                             | 8 | 3 |   |
|                 | Potassium (K)  | 19            | 2                             | 8 | 8 | 1 |
|                 | Calcium (Ca)   | 20            | 2                             | 8 | 8 | 2 |
| Non-metals      | Nitrogen (N)   | 7             | 2                             | 5 |   |   |
|                 | Oxygen (O)     | 8             | 2                             | 6 |   |   |
|                 | Fluorine (F)   | 9             | 2                             | 7 |   |   |
|                 | Phosphorus (P) | 15            | 2                             | 8 | 5 |   |
|                 | Sulphur (S)    | 16            | 2                             | 8 | 6 |   |
|                 | Chlorine (Cl)  | 17            | 2                             | 8 | 7 |   |





# PROPERTIES OF IONIC COMPOUNDS

- **Physical nature**: generally solid & hard (strong electrostatic force between ions); brittle
- **M.pt. & B.pt**: generally ↑ due to strong interionic attraction.
- **Soluble in water** & insoluble in petrol & kerosene
- **Conduction of electricity**: do not conduct in solid state; conduct in molten & aqueous state

# ALLOY

- **Homogenous mixture of 2 or more metals or a metal & non-metal**
- Initially, a metal is melted, then another element is dissolved in definite proportion & then cooled

|                         |                                                        |
|-------------------------|--------------------------------------------------------|
| Amalgam                 | Hg Alloy                                               |
| Brass (poor conductor)  | Cu and Zn                                              |
| Bronze (poor conductor) | Cu, Sn (Tin)                                           |
| Solder (low m.pt)       | Pb, Sn                                                 |
| Alloy of Au             | Ag and Cu are added in 24 carat gold to make jewellery |
| Steel                   | Fe, Ni, Cr (No rust, hard)                             |

- **Properties**: stronger, harder, resistant to corrosion, Low m.pt., electrical conductivity ↑

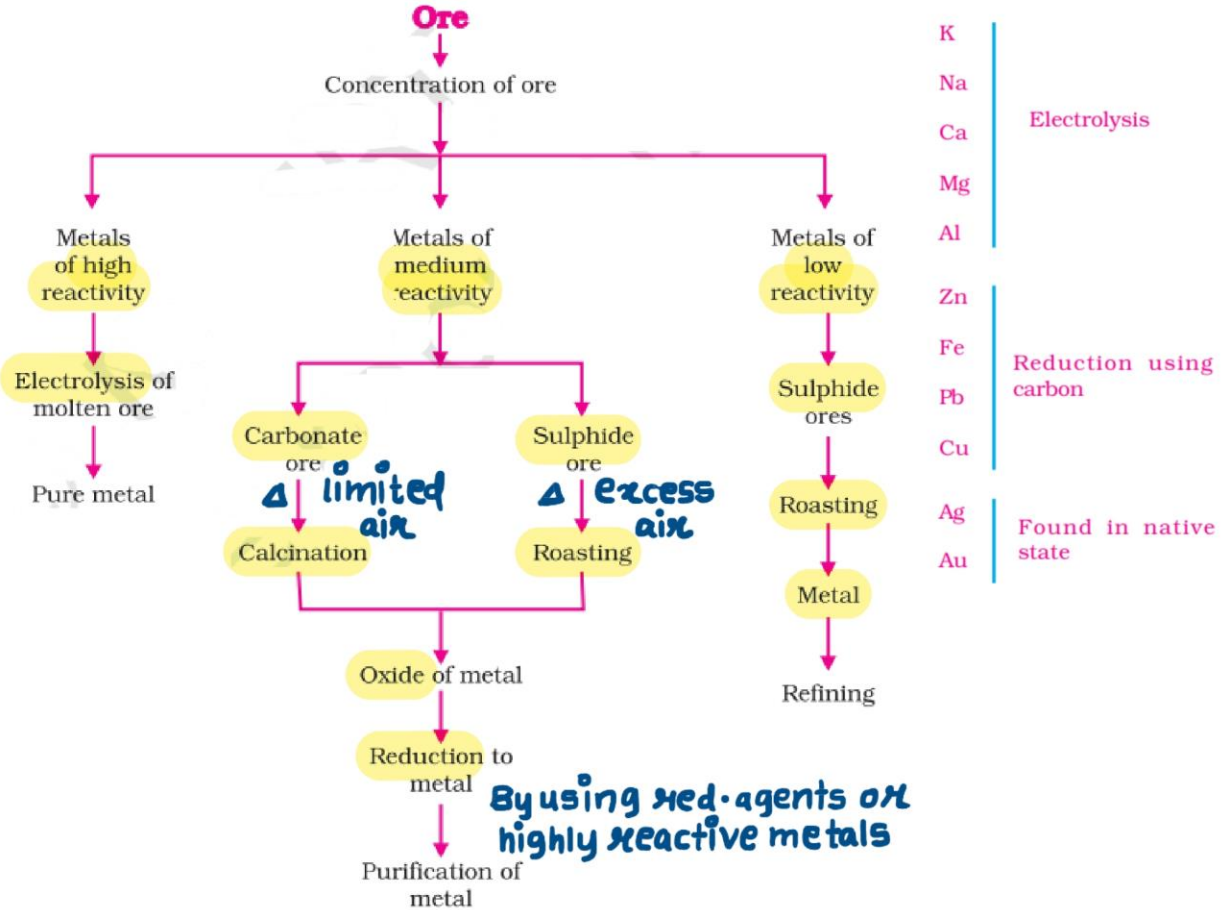
# OCCURRENCE

- **Minerals**: occur naturally in Earth's crust
- **Ores**: Minerals from which a metal can be profitably extracted
- sea water contains soluble salts: NaCl, MgCl<sub>2</sub> etc.
- ores are mainly oxides as oxygen is highly reactive & abundant

|                                                                                                                                                  |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Most reactive metals</b> <ul style="list-style-type: none"> <li>• Always found in combined state</li> </ul>                                   | K, Na, Ca, Mg, Al  |
| <b>Moderately reactive metals</b> <ul style="list-style-type: none"> <li>• Found in earth's crust as sulphides, oxides and carbonates</li> </ul> | Zn, Fe, Pb         |
| <b>Least reactive metals</b> <ul style="list-style-type: none"> <li>• Generally found in free state</li> </ul>                                   | Cu, Hg, Ag, Au, Pt |

- **Gangue**: impurities, sand, soil, dust; separated from main metal due to diff. in physical & chemical properties
- Diff. separation techniques are employed based on type of gangue





## EXTRACTION OF METALS

### 1 Metals with high reactivity

- oxides can't be reduced by 'C' ∴ metals have high affinity for 'O' than 'C'
- ∴ **Electrolytic reduction** needed
- Na, Ca, Mg: Electrolysis of their molten chlorides
- Al: Electrolysis of its oxide

e.g., NaCl: At anode ⊕ve:  $\text{Cl}_2$

At cathode ⊖ve:  $\text{Na}$

### 2 Metals with Medium reactivity:

Sulphides/Carbonates  $\xrightarrow{\Delta}$  oxides  $\xrightarrow{\text{Reduced}}$  Metal

Easier to reduce

• Roasting:  $2\text{ZnS (s)} + 3\text{O}_2 \text{ (g)} \xrightarrow{\Delta} 2\text{ZnO (s)} + 2\text{SO}_2 \text{ (g)}$

• Calcination:  $\text{ZnCO}_3 \text{ (s)} \xrightarrow{\Delta} \text{ZnO (s)} + \text{CO}_2 \text{ (g)}$

• Reduction: by suitable reducing agents

$\text{ZnO (s)} + \text{C (s)} \longrightarrow \text{Zn (s)} + \text{CO (g)}$

• Reduction: by using highly reactive metal

$3\text{MnO}_2 \text{ (s)} + 4\text{Al (s)} \longrightarrow 3\text{Mn (l)} + 2\text{Al}_2\text{O}_3 \text{ (s)} + \Delta$   
↳ Molten ∴ exothermic

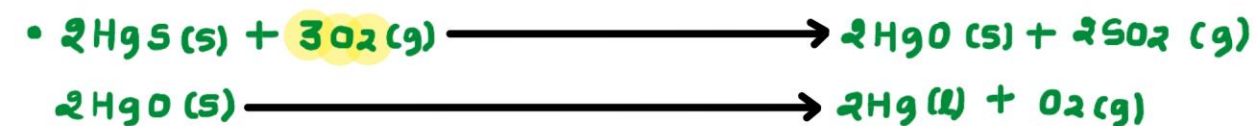
• Thermite reaction:

$\text{Fe}_2\text{O}_3 \text{ (s)} + 2\text{Al (s)} \longrightarrow 2\text{Fe (l)} + \text{Al}_2\text{O}_3 \text{ (s)} + \Delta$

### 3 Metals with low reactivity:

sulphides  $\xrightarrow{\Delta}$  oxides  $\xrightarrow{\Delta}$  Metals

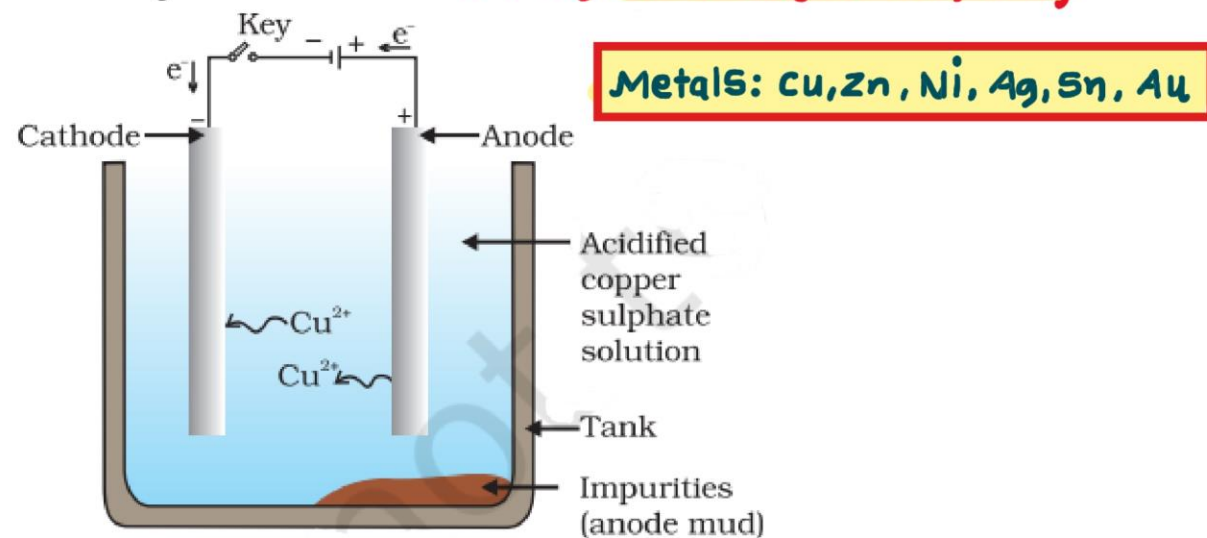
• HgS (cinnabar)



Reaction mixture

## REFINING OF METALS

- Removal of impurities from extracted metal to obtain pure metal e.g., By electrolytic refining



- Amount of pure metal = Amount of electrolyte used



# CARBON AND ITS COMPOUNDS

## COVALENT BONDING

- Bonding done by sharing of  $e^-$  pairs to fill their outermost shell (to attain noble gas configuration  $\therefore$  stability)
- Carbon has atomic no. 6  $\therefore$  electronic configuration = 2, 4

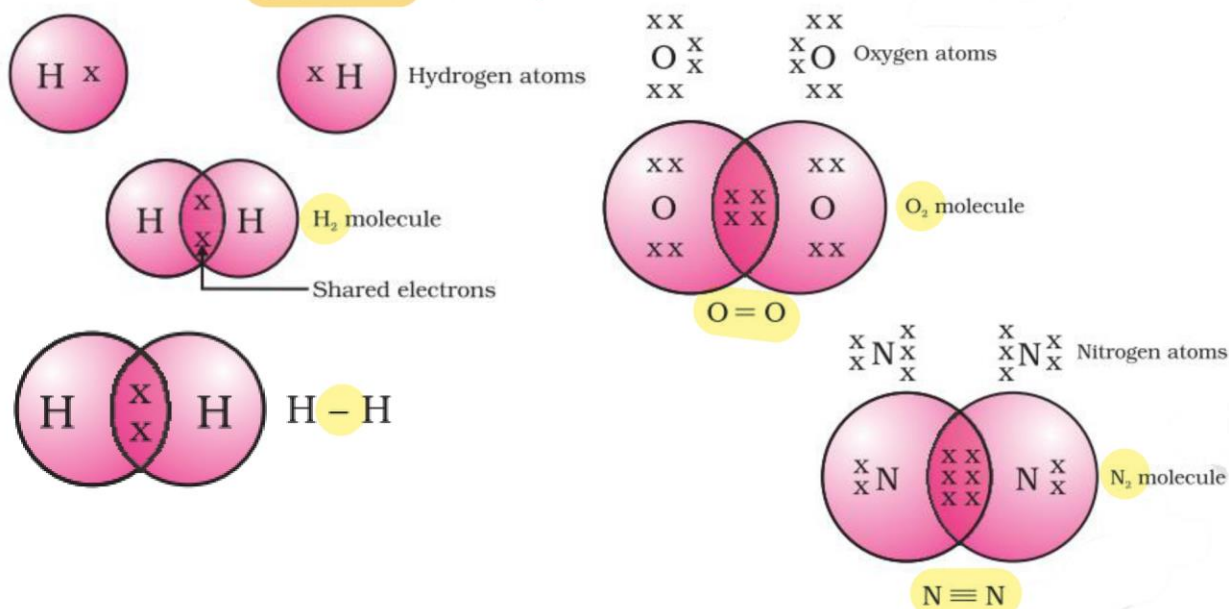
**Case-I: carbon can gain 4 electrons  $\therefore C^{4-}$**

It is difficult for 6 protons to hold 10 electrons

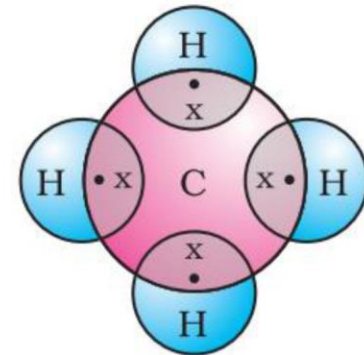
**Case-II: carbon can lose 4 electrons  $\therefore C^{4+}$**

Again difficult for 2 electrons to be with 6 protons

$\therefore$  carbon shares valence electrons



- Such bonds are formed between non-metals e.g.,  $N_2$ ,  $O_2$ ,  $H_2O$ ,  $Cl_2$ ,  $CH_4$

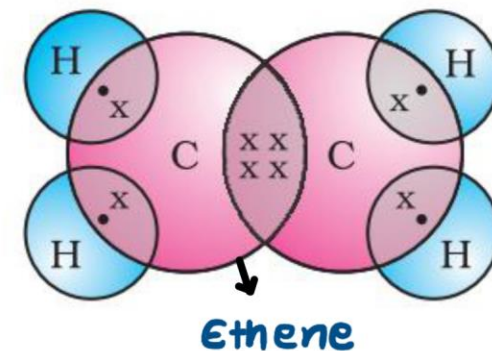
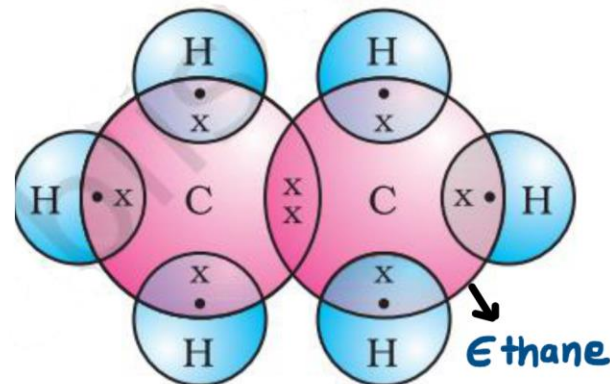


## Properties of covalent compounds

- Poor conductors (free  $e^-$ , ions  $\ominus$ nt); low m.pt. or b.pt due to weak intermolecular forces

## ELECTRON DOT STRUCTURE/ LEWIS STR.

- Provide information about shared electron in a molecule
- electrons are represented by dots & crosses

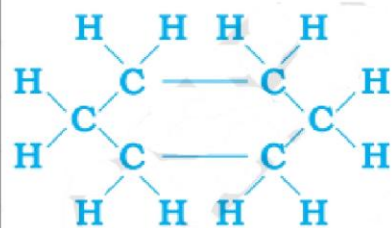


## NOMENCLATURE OF ORG. COMPOUNDS

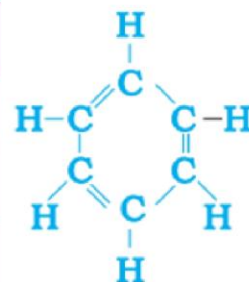
- Was done on the basis of no. of carbons; presence of saturation, unsaturation or functional group



| No. of carbon Atoms | Prefix | Nature/Type of C-C bond | Primary Suffix     |
|---------------------|--------|-------------------------|--------------------|
| 1                   | Meth-  | C-C                     | -ane $C_nH_{2n+2}$ |
| 2                   | Eth-   | C=C                     | -ene $C_nH_{2n}$   |
| 3                   | Prop-  |                         |                    |
| 4                   | But-   | C≡C                     | -yne $C_nH_{2n-2}$ |
| 5                   | Pent-  |                         |                    |
| 6                   | Hex-   |                         |                    |
| 7                   | Hept-  |                         |                    |
| 8                   | Oct-   |                         |                    |
| 9                   | Non-   |                         |                    |



cyclo-hexane



Benzene

| Class of compounds         | Prefix/Suffix              | Example                                                                                                                                                                                    |
|----------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Halo alkane<br>—Cl, —Br | Prefix-chloro, bromo, etc. | $\begin{array}{c} \text{H} & \text{H} & \text{H} \\   &   &   \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{Cl} \\   &   &   \\ \text{H} & \text{H} & \text{H} \end{array}$ Chloropropane   |
|                            |                            | $\begin{array}{c} \text{H} & \text{H} & \text{H} \\   &   &   \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{Br} \\   &   &   \\ \text{H} & \text{H} & \text{H} \end{array}$ Bromopropane    |
| 2. Alcohol —OH             | Suffix -ol                 | $\begin{array}{c} \text{H} & \text{H} & \text{H} \\   &   &   \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{OH} \\   &   &   \\ \text{H} & \text{H} & \text{H} \end{array}$ Propanol        |
| 3. Aldehyde                | Suffix -al                 | $\begin{array}{c} \text{H} & \text{H} & \text{H} \\   &   &   \\ \text{H}-\text{C}-\text{C}-\text{C}=\text{O} \\   &   &   \\ \text{H} & \text{H} & \text{H} \end{array}$ Propanal         |
| 4. Ketone                  | Suffix -one                | $\begin{array}{c} \text{H} & & \text{H} \\   & &   \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\   &    &   \\ \text{H} & \text{O} & \text{H} \end{array}$ Propanone                  |
| 5. Carboxylic acid         | Suffix -oic acid           | $\begin{array}{c} \text{H} & \text{H} & \text{O} \\   &   &    \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{OH} \\   &   &   \\ \text{H} & \text{H} & \text{H} \end{array}$ Propanoic acid |

## SPECIAL NATURE OF CARBON

- Catenation:** Self linking property of carbon to make long chains; due to its small size and strong C-C bonds (stability↑)
  - It can make straight chain, branched chain & cyclic compounds
- Tetravalent nature:** carbon's valency is 4. It can make bonds with 4 other carbons or heteroatoms like O, H, Cl, N etc.
- can make different compounds:**
  - Saturated:** Alkane (—)
  - Unsaturated:** Alkene (=) & Alkyne (≡)
  - Aromatic:** have alternate single & double bonds: Benzene

## HOMOLOGOUS SERIES

- series of organic comp. having same functional group (∴ same chemical properties); successive members differ by a  $-\text{CH}_2$  unit or 14 amu mass
- B.pt↑ and M.pt↑ as mass increases



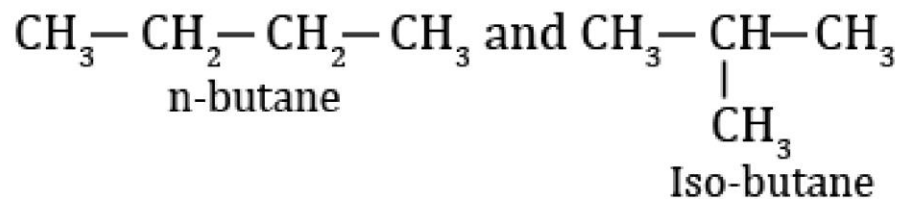
e.g., Ethanol:  $\text{CH}_3\text{CH}_2\text{OH}$

Propanol:  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

Butanol:  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

## ISOMERS

- Isomers have same molecular formula but diff. structure

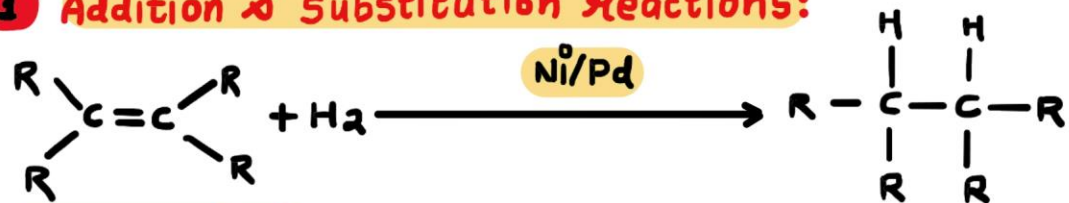


- Functional isomers: same formula but diff. funct. groups

Propanal ( $\text{C}_3\text{H}_6\text{O}$ ), Propanone ( $\text{C}_3\text{H}_6\text{O}$ )

## CHEMICAL PROPERTIES

- Addition & substitution reactions:



Unsaturated

e.g., use to make vegetable ghee from vegetable oil



Saturated

## 2 Combustion:

Carbon compound +  $\text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Heat} + \text{light}$

$\text{C} + \text{O}_2 \longrightarrow \text{CO}_2 + \text{Heat}$

$\text{CH}_4 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Heat}$

$\text{CH}_3\text{CH}_2\text{OH} + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Heat}$

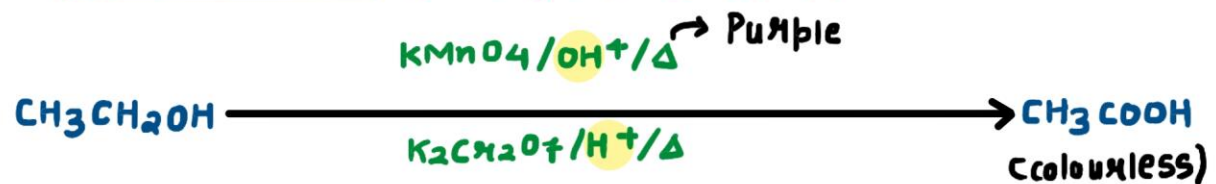
- Saturated: clean blue flame

- Unsaturated: yellow flame with black sooty smoke

↪ Naphthalene/camphor

- Coal & Petrol:  $\text{CO}_2$  + oxides of 'N' and 'S' (pollutants)

## 3 Oxidation by Pot. Permanganate (alk. $\text{KMnO}_4$ ) or Pot. dichromate (acidified $\text{K}_2\text{Cr}_2\text{O}_7$ )



## REACTIONS OF ETHANOL/ ALCOHOL

- Properties: liq. at room temp.; soluble in  $\text{H}_2\text{O}$ ; methanol caused blindness; absolute alcohol is lethal



- Dehydration:

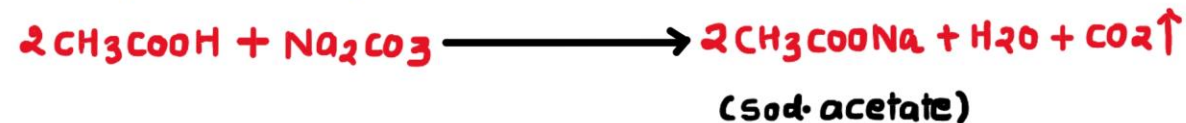


## REACTIONS OF ETHANOIC ACID/ ACETIC ACID

- 5-8% acetic acid in  $\text{H}_2\text{O}$  is vinegar (preservative); freezes in winter  $\therefore$  **glacial acetic acid** (m.pt.  $17^\circ\text{C}$ ); weak acid

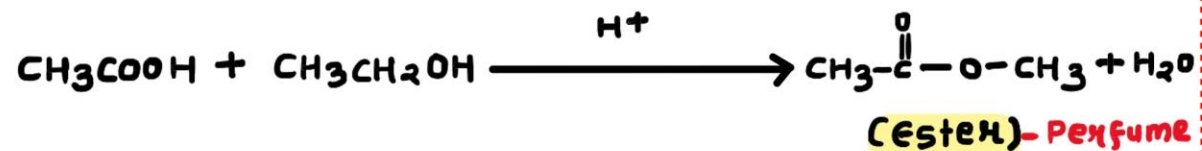


- Test for carboxylic acid



- Esterification/saponification

$\hookrightarrow$  Acid + Alcohol  $\hookrightarrow$  Ester + NaOH



- Soaps are Na & K salts of long chain of acids

## CLEANSING ACTION OF SOAP

- **Hard  $\text{H}_2\text{O}$** : do not produce lather with soap due to presence of 'Ca & Mg' salts (sulphates, chlorides, bicarbonate)

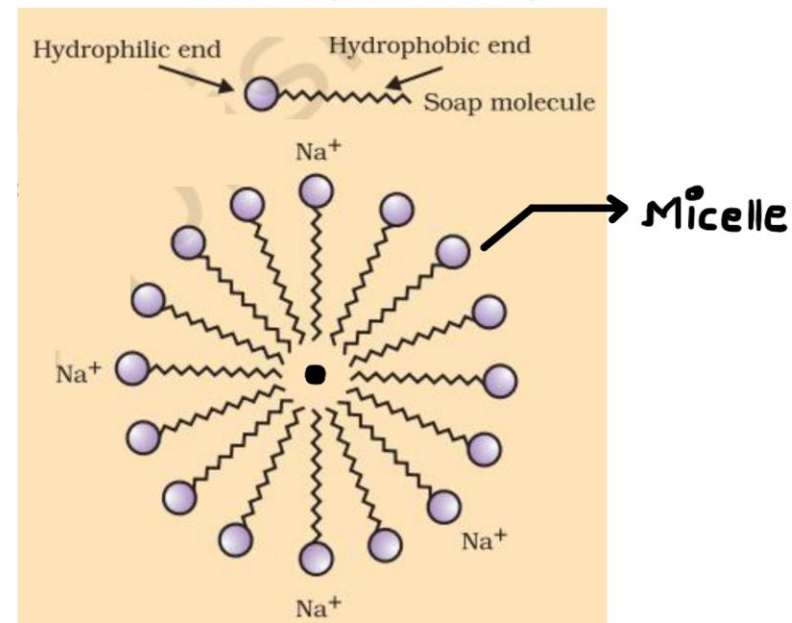
- **Soft  $\text{H}_2\text{O}$** : free from 'Ca & Mg' salts  $\therefore$  produce lather

- **Soaps**:  $\text{RCOO}^-\text{Na}^+$  (R = 15 to 17 carbon long chain)

Make scum by reacting with Ca & Mg

- **Detergents**: ammonium & sulphonate salts of long chain fatty acids —  $\text{OSO}_3^-\text{Na}^+$

$\hookrightarrow$  do not make scum





# ALLOTROPE OF CARBON

- carbon occurs in various forms with different physical yet similar chemical properties

1 **Fullerenes:**  $C_{60}$  ; football like

2 **Graphite:** every carbon is arranged with 3 other carbons  $\therefore$  hexagonal array

- conductor, smooth & slippery, used in lead pencil

3 **Diamond:** each 'C' is attached to 4 other 'C' forming rigid 3-D structure (hardest known substance)